

**SAMPLE FORM – NOT VALID FOR SUBMISSION TO BUILDING DEPARTMENTS****CERTIFICATE OF VERIFICATION****Note:** This table completed by **HERSECC** Registry.

Project Name:	Enforcement Agency:
Project Address:	Permit Number:
City and Zip Code:	Permit Application Date:

Title 24, Part 6, Section 160.2(b)2 **Ventilation and Indoor Air Quality for Attached Dwelling Units**. All dwelling units shall meet the requirements of ANSI/ASHRAE Standard 62.2-~~2019~~-2022 Ventilation and Acceptable Indoor Air Quality in Residential Buildings subject to the amendments specified by Title 24, Part 6, Section 160.2(b)2A.

A. Whole-Dwelling Mechanical Ventilation - General Information

01	Dwelling Unit Name	
02	Building Type	
03	Project Scope	
04	Total Conditioned Floor Area of Dwelling Unit (For addition projects the conditioned floor area equals existing area plus addition area)	
05	Number of Bedrooms in Dwelling Unit (For addition projects the number of bedrooms equals the existing bedrooms plus addition bedrooms)	
06	Ventilation System Type	
07	Ventilation Operation Schedule	
08	Fault Indicator Display (FID) Status	

Note:**MCH-27 – High-Rise Multifamily Ventilation****B. Ventilation - Total Ventilation Rate** ~~MCH 27b – High-rise Residential Multifamily Ventilation~~

A mechanical supply system, exhaust system, or combination thereof shall provide whole-dwelling ventilation with outdoor air each hour at no less than the rate in 160.2(b)2Aiv

01	Total Required Ventilation rate, (Q_{tot})	
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C. Installed Ventilation - Total Ventilation Rate

A mechanical supply system, exhaust system, or combination thereof shall provide whole-dwelling ventilation with outdoor air each hour at no less than the rate in 160.2(b)2Aiv

01	02	03	04	05
Fan Name	Fan Location	Runtime (Min/Hr)	Installed Mechanical Ventilation Rate (CFM)	Equivalent Continuous Ventilation (CFM)
06	Total Installed Equivalent Continuous Ventilation (CFM)			

D. Additional Envelope Requirements

01	Envelope Leakage	
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E. Additional Central Ventilation System Balancing Requirements

01	Maximum Ventilation Flow (CFM)	
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**SAMPLE FORM – NOT VALID FOR SUBMISSION TO BUILDING DEPARTMENTS****F. HRV or ERV serving Individual Dwelling Unit**

- Heat or Energy Recovery Systems must have a fan efficacy of ≤ 1.0 W/cfm in all climate zones (Section 160.2(b)2Biii).
- Heat or Energy Recovery Systems must prescriptively have a fan efficacy of ≤ 0.6 W/cfm and a minimum sensible heat recovery of 67% in climate zones 1, 2, 4 and 11-14 and 16 (Section 170.2(c)3Biva).

01	02	03	04
Manufacturer Make	Manufacturer Model Number	Fan Efficacy Performance Rating (W/CFM)	Sensible Recovery Efficiency (%)

G. Requirements for balanced and supply only ventilation systems (160.0(b)2Axi)

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.

Balanced and supply ventilation component accessibility. Balanced and supply ventilation systems shall meet the following requirements for accessibility:

01	IAQ filter and HRV/ERV accessibility. System air filters and HRV/ERV heat/energy recovery cores shall be located such that they are accessible for service from within occupiable spaces, basements, garages, balconies, mechanical closets or accessible rooftops. Filters and heat/energy recovery cores behind access panels, access doors, or grilles located no more than 10 feet above a walking surface inside a space specified above comply with this requirement. Exception to Section 160.2(b)2Axi: Systems that require servicing from inside the attic shall have the following: 1. A Fault Indicator Display (FID) meeting the requirements of Reference Appendix JA17; and 2. An attic access door located in a wall or, where attic access is provided through a ceiling, an attic access hatch that includes an integrated ladder; and 3. A walkway from the attic access door to the HRV/ERV.
02	IAQ system component accessibility. Fans, motors, heat exchangers, filters and recovery cores shall meet all applicable requirements of California Mechanical Code 304.0 accessibility of service.

H. Fault Indicator Display Installation Verification

Procedures for the Fault Indicator Display Verification are detailed in in Appendix JA17.

01	FID Manufacturer Name/Make	
02	FID Model Number	
03	The display module is mounted adjacent to the system thermostat.	
04	The manufacturer has certified to the Energy Commission that the FID model meets the requirements of Reference Joint Appendix JA17 (make and model found on CEC list of approved FID devices).	
05	The system has operated for at least 15 minutes and the FID reports that the system is operating within acceptable parameters.	
06	Compliance Statement:	

**SAMPLE FORM – NOT VALID FOR SUBMISSION TO BUILDING DEPARTMENTS****I. Fault Indicator Display – Additional Requirements**

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.

01	Fault Indicator Display devices shall either be factory installed by the space-conditioning system manufacturer, or field installed according to the space-conditioning system manufacturer's requirements and the FID manufacturer's specifications.
02	The installer shall ensure that a copy of the FID manufacturer's user instructions documentation has been made available to the building owner.

FJ. Compliance Statement

01	
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GK. Determination of ~~HERS~~ECC Verification Compliance

All applicable sections of this document shall indicate compliance with the specified verification protocol requirements in order for this Certificate of Verification as a whole to be determined to be in compliance

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**SAMPLE FORM – NOT VALID FOR SUBMISSION TO BUILDING DEPARTMENTS**

DOCUMENTATION AUTHOR'S DECLARATION STATEMENT	
1. I certify that this Certificate of Verification documentation is accurate and complete.	
Name:	Signature:
Company:	Date:
Address:	CEA / HERSECC Certification Identification (if applicable):
City/State/Zip:	Phone:
RESPONSIBLE PERSON'S DECLARATION STATEMENT	
I certify the following under penalty of perjury, under the laws of the State of California:	
1. The information provided on this Certificate of Verification is true and correct. 2. I am the certified HERSECC Rater who performed the verification identified and reported on this Certificate of Verification (responsible rater). 3. The installed features, materials, components, manufactured devices, or system performance diagnostic results that require HERSECC verification identified on this Certificate of Verification comply with the applicable requirements in Reference Nonresidential Appendices NA12 and NA2, NA7 and the requirements specified on the Certificate of Compliance for the building approved by the enforcement agency. 4. The information reported on applicable sections of the Certificate(s) of Installation (NRCI), signed and submitted by the person(s) responsible for the construction or installation conforms to the requirements specified on the Certificate(s) of Compliance (NRCC) approved by the enforcement agency. 5. I understand that a registered copy of this Certificate of Verification shall be posted, or made available with the building permit(s) issued for the building, and made available to the enforcement agency for all applicable inspections, and I will take the necessary steps to accomplish this requirement. 6. I understand that a registered copy of this Certificate of Verification is required to be included with the documentation the builder provides to the building owner at occupancy, and I will take the necessary steps to accomplish this requirement.	
BUILDER OR INSTALLER INFORMATION AS SHOWN ON THE CERTIFICATE OF INSTALLATION	
Company Name (Installing Subcontractor or General Contractor or Builder/Owner):	
Responsible Builder/Installer Name:	CSLB License:
HERSECC PROVIDER DATA REGISTRY INFORMATION	
Sample Group Number (if applicable):	Dwelling Test Status in Sample Group (if applicable)
HERSECC RATER INFORMATION	
HERSECC Rater Company Name:	
Responsible Rater's Name:	Responsible Rater's Signature:
Responsible Rater's Certification Number w/ this HERSECC Provider	Date Signed:

NRCV-MCH-27b-H User Instructions

Section A. General Information

1. Dwelling Unit Name: User input text from the NRCC-PRF-01 (Performance) or NRCC-MCH-~~E01~~ (Prescriptive). This is the unique identifier for this dwelling unit. Needed for high-rise residential dwelling units. Ventilation is calculated and provided for each dwelling unit individually.
2. Building Type: Fixed value equal to high-rise residential.
3. Project Scope: User select from following new, addition, or alteration. Based on project scope from the NRCC-PRF-01 (Performance) or NRCC-MCH-~~E01~~ (Prescriptive).
4. Total Conditioned Floor Area of Dwelling Unit: User input number based on the information from NRCC-PRF-01 (Performance) or NRCC-MCH-~~E01~~ (Prescriptive).
5. Number of Bedrooms in Dwelling Unit: User input number based on the information from NRCC-PRF-01 (Performance) or NRCC-MCH-~~E01~~ (Prescriptive).
6. Ventilation system Type: This is a user selected value from list of ventilation types Supply, Exhaust, Balanced, Balanced – ERV, Balanced – HRV, Central Fan Integrated (CFI), Central Ventilation System – Supply and Central Ventilation System – Exhaust and Central Ventilation System Balanced.
7. Ventilation operation schedule: This is a user selected value from list of Continuous, Short-Term Average, Scheduled and Real-time Control.
 - Note if “Ventilation System Type” (A06) = Central Fan Integrated & “Ventilation Operation Schedule” (A07) = Continuous; then user will not be allowed to proceed.

Section B. Ventilation - Total Ventilation Rate –~~High-rise Residential Multifamily Ventilation~~

This value is automatically calculated using equation 160.2-B from the Energy Standards.

Section C. Installed Ventilation – Total Ventilation Rate Method

1. User input text identifying the fan name for each installed ventilation fan.
2. User input text identifying the fan location for each installed ventilation fan.
3. Runtime (Min/Hr): This value may be filled out automatically or be user input.
 - If ventilation operation schedule from section A = “continuous”, then value of 60 will be automatically entered.
 - If ventilation operation schedule from section A = “short term average”, then user enter value of less than or equal to 60 for each installed ventilation fan.
4. User to enter CFM value from test procedures described in NA7.18.1 for each installed ventilation fan.
5. Equivalent continuous ventilation CFM is automatically calculated for each ventilation fan.
6. Total installed equivalent continuous ventilation CFM is automatically calculated based on the installed ventilation fans.

Section D. HRV or ERV serving Individual Dwelling Unit

1. User input manufacturer make of the installed equipment from the manufacturer nameplate.
2. User input model number of the installed equipment from the manufacturer nameplate.

3. User input the fan efficacy performance rating (W/CFM) for the installed equipment as determined by RA3.7.4.4.
4. User input the sensible recovery efficiency performance rating (%) for the installed equipment as determined by RA3.7.4.4.

Section DE. Additional Envelope Requirements

1. Envelope Leakage: This field is filled out automatically. It is referenced from the NRCC-MCH-24, which must be completed prior to this document.

Section EF. Additional Central Ventilation System Balancing Requirements

1. Maximum Ventilation Flow (CFM): This field is filled out automatically calculated.

Section G. Requirements for balanced and supply only ventilation systems

1. This statement must be true (or not applicable) for the test to conform to the protocols.
2. This statement must be true (or not applicable) for the test to conform to the protocols.

Section H. Fault Indicator Display

1. Enter the manufacturer name or make of the approved Fault Indicator Display. Must match name shown on the list of approved devices kept by the Commission.
2. Enter the manufacturer model number of the approved Fault Indicator Display. Must match name shown on the list of approved devices kept by the Commission.
3. The installer must confirm that the FID display module is mounted adjacent to thermostat that controls the system being verified. This requirement is detailed in Residential Appendix JA17.
4. The installer must confirm that the installed FID is approved and appears the list of approved devices kept by the Commission. This requirement is detailed in Residential Appendix JA17.
5. The installer must confirm that the system has operated for at least 15 minutes and that they system is operating within acceptable parameters as specified by the FID and equipment manufacturers. This requirement is detailed in Residential Appendix JA17.

Section I. Fault Indicator Display – Additional Requirements

1. This statement must be true (or not applicable) for the test to conform to the protocols.
2. This statement must be true (or not applicable) for the test to conform to the protocols.

Section J. Compliance Statement

1. Compliance Statement: This field is filled out automatically

Section K. Determination of ECC Verification Compliance

1. This field is filled out automatically based on all verification protocol requirements in this document showing compliance.

Title 24, Part 6, Section 160.2(b)2 **Ventilation and Indoor Air Quality for Attached Dwelling Units.** All dwelling units shall meet the requirements of ANSI/ASHRAE Standard 62.2-2019 Ventilation and Acceptable Indoor Air Quality in Residential Buildings subject to the amendments specified by Title 24, Part 6, Section 160.2(b)2A.

A. Whole-Dwelling Mechanical Ventilation - General Information

01	Dwelling Unit Name	<<user input, text, maximum 50 characters>>
02	Building Type	<< Fixed value "High-rise Residential">>
03	Project Scope	<<user pick select from list: New; Addition; or Alteration>>
04	Total Conditioned Floor Area of Dwelling Unit (For addition projects the conditioned floor area equals existing area plus addition area)	<<user input, number, integer XXX>>
05	Number of Bedrooms in Dwelling Unit (For addition projects the number of bedrooms equals the existing bedrooms plus addition bedrooms)	<<user input, number, integer XXX>>
06	Ventilation System Type	<<Calculated value, allow user pick one from list: **Supply **Exhaust; or **Balanced; or **Balanced – ERV; or **Balanced – HRV; or **Central Fan Integrated (CFI); or **Central Ventilation System – Supply; or **Central Ventilation System – Exhaust; or **Central Ventilation System – Balanced >>
07	Ventilation Operation Schedule	<< Calculated value, allow user pick one from list: **Continuous; or **Short-Term Average; Else if "Ventilation System Type" (A06) = Central Fan Integrated & "Ventilation Operation Schedule" (A07)= Continuous; then display: "Central Fan Integrated Ventilation System Not Allowed to Operate Continuously - Do Not Proceed ">>
08	<u>Fault Indicator Display (FID) Status</u>	<<auto filled text: referenced from LMCI. Possible entries are "This system has a factory installed FID"; or "This system has a field installed FID"; or "This system does not have a FID device installed" If installed system does not match this entry, it can be overwritten by rater but it will be flagged as a possible fail>>>
09	determine compliance method for this document; display applicable tables below; (this row not visible to user)	<<calculated field: Else if "Building Type" (A02) = High-rise Residential and "Ventilation Operation Schedule" (A07) ≠ Scheduled or Real-Time Control, then display method: **27b – High-rise Residential Ventilation; >>

Note:

Non-dwelling units do not meet the definition for a dwelling unit as defined in Section 100.1(b). Non-dwelling units are not designed to provide independent living facilities and do not provide permanent provisions for living, sleeping, eating, cooking and sanitation.

MCH-27b – High-rise Multifamily Ventilation

B. Ventilation - Total Ventilation Rate - MCH-27b – High-rise Residential Multifamily Ventilation

A mechanical supply system, exhaust system, or combination thereof shall provide whole-dwelling ventilation with outdoor air each hour at no less than the rate in 160.2(b)2Aiv.

01	Total Required Ventilation rate, (Q_{tot})	<<calculated field, numeric: (use equation 160.2-B): $[(0.03 * \text{"Total Conditioned Floor Area of Dwelling Unit"} (A04)) + (7.5 * (\text{"Number of Bedrooms in Dwelling Unit"} (A05) + 1))]$, (cfm). >>
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C. Installed Ventilation - Total Ventilation Rate

A mechanical supply system, exhaust system, or combination thereof shall provide whole-building ventilation with outdoor air each hour at no less than the rate in 160.2(b)2Aiv.

01	02	03	04	05
Fan Name	Fan Location	Runtime (Min/Hr)	Installed Mechanical Ventilation Rate (CFM)	Equivalent Continuous Ventilation (CFM)
<< user input, text>>	<<user input, text>>	<<calculated field: if value in "Ventilation Operation Schedule" (A07) equals Continuous, then value equals 60;	<< user input, positive integer>>	<<calculated field, value = ("Runtime (Min/Hr)" (C03) * "Installed Mechanical Ventilation Rate (CFM)" (C04)) / 60 (CFM)>>
		Else if value in "Ventilation Operation Schedule" (A07) equals Short Term Average, then user input value positive integer ≤ 60>>		
06	Total Installed Equivalent Continuous Ventilation (CFM)			<<calculated field, value = sum of values in column "Equivalent Continuous Ventilation (CFM)" (C05)>>

D. HRV or ERV serving Individual Dwelling Unit

- Heat or Energy Recovery Systems must have a fan efficacy of ≤ 1.0 W/cfm in all climate zones (Section 160.2(b)2Biii).
- Heat or Energy Recovery Systems must prescriptively have a fan efficacy of ≤ 0.6 W/cfm and a minimum sensible heat recovery of 67% in climate zones 1, 2, 4, and 11-14, and 16 (Section 170.2(c)3Biva).

01	02	03	04
Manufacturer Make	Manufacturer Model Number	Fan Efficacy Performance Rating (W/CFM)	Sensible Recovery Efficiency (%)
<< user input, text>>	<< user input, text>>	<< user input, text>>	<< user input, text>>
		Range check: User entered value must be ≤ 0.6 if complying prescriptively in climate zones 1, 2, 4, 11-14 and 16; else value must be ≤ 1.0>>	Range check: User entered value must be ≥ 67% if complying prescriptively in climate zones 1, 2, 4, 11-14, and 16; else no restriction>>

DE. Additional Envelope Requirements

<<if Ventilation System Type (A06) equals Supply, Exhaust, Central Fan Integrated (CFI), Central Ventilation System – Supply or Central Ventilation System – Exhaust then display Table D; Else display the section does not apply message>>

01	Envelope Leakage	<< calculated field, referenced data from NRCV-MCH-24>>
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EF. Additional Central Ventilation System Balancing Requirements

<<if Ventilation System Type (A06) equals Central Ventilation System – Supply, Central Ventilation System – Exhaust or Central Ventilation System – Balanced then display Table E; Else display the section does not apply message>>

01	Maximum Ventilation Flow (CFM)	<<calculated field, "Required Mechanical Ventilation Rate (B01) * 1.20>>
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G. Requirements for balanced and supply only ventilation systems (160.0(b)2Axi)

<<If A06=Balanced HRV or Balanced ERV then display table; Else display the "section does not apply" message>>

<<Balanced and supply ventilation component accessibility. Balanced and supply ventilation systems shall meet the following requirements for accessibility:>>

01	IAQ filter and HRV/ERV accessibility. System air filters and HRV/ERV heat/energy recovery cores shall be located such that they are accessible for service from within occupiable spaces, basements, garages, balconies, mechanical closets or accessible rooftops. Filters and heat/energy recovery cores behind access panels, access doors, or grilles located no more than 10 feet above a walking surface inside a space specified above comply with this requirement. Exception to Section 160.2(b)2Axi: Systems that require servicing from inside the attic shall have the following: 1. A Fault Indicator Display (FID) meeting the requirements of Reference Appendix JA17; and 2. An attic access door located in a wall or, where attic access is provided through a ceiling, an attic access hatch that includes an integrated ladder; and 3. A walkway from the attic access door to the HRV/ERV.
02	IAQ System component accessibility: Fans, motors, heat exchangers, filters and recovery cores shall meet all applicable requirements of California Mechanical Code Section 304.0 accessibility for service.
The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.	

H. Fault Indicator Display Installation Verification

<<Procedures for the Fault Indicator Display Verification are detailed in in Appendix JA17>>

01	FID Manufacturer Name/Make	<<auto filled text: referenced from LMCI. If installed system does not match this entry, it can be overwritten by rater but it will be flagged as a possible fail.>>
02	FID Model Number	<<auto filled text: referenced from LMCI. If installed system does not match this entry, it can be overwritten by rater but it will be flagged as a possible fail.>>
03	The display module is mounted adjacent to the system thermostat.	<<user entry, select from choices "yes" or "no">>
04	The manufacturer has certified to the Energy Commission that the FID model meets the requirements of Reference Joint Appendix JA17 (make and model found on CEC list of approved FID devices).	<<user entry, select from choices "yes" or "no">>
05	The system has operated for at least 15 minutes and the FID reports that the system is operating within acceptable parameters.	<<user entry, select from choices "yes" or "no">>
06	Compliance Statement:	<<If A03, A04 and A05 = "yes", then print statement, "FID complies", Else print "FID does NOT comply">>

I. Fault Indicator Display – Additional Requirements

<<If A08 = "This system has a factory installed FID"; or "This system has a field installed FID", then display this section>>

<<The responsible persons signature on this document indicates the installation complies with the following requirements:>>

01	Fault Indicator Display devices shall either be factory installed by the space-conditioning system manufacturer, or field installed according to the space-conditioning system manufacturer's requirements and the FID manufacturer's specifications.
02	The installer shall ensure that a copy of the FID manufacturer's user instructions documentation has been made available to the building owner.

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.

FJ. Compliance Statement

01	<<If Else if the “Ventilation System Type” (A06) equals Balanced, Balanced – ERV or Balanced – HRV and the “Total Installed Equivalent Continuous Ventilation” (C06) $\geq$ “Total Required Ventilation Rate” (B01), then display text: “Building Passes Mechanical Ventilation Rate Test” Else if the “Ventilation System Type” (A06) equals Central Ventilation System – Balanced and the “Total Installed Equivalent Continuous Ventilation” (C06) $\geq$ “Total Required Ventilation Rate” (B01) and the “Total Installed Equivalent Continuous Ventilation” (C06) $\leq$ “Maximum Ventilation Flow” (E01), then display text: “Building Passes Mechanical Ventilation Rate Test” Else if the “Ventilation System Type” (A06) equals Supply, Exhaust, Central Fan Integrated (CFI) and the “Total Installed Equivalent Continuous Ventilation” (C06) $\geq$ “Total Required Ventilation Rate” (B01), and if “Envelope Leakage” (D01) $\leq$ “Target dwelling unit compartmentalization leakage” (taken from NRCV-MCH-24), then display text: “Building Passes Mechanical Ventilation Rate Test” Else if the “Ventilation System Type” (A06) equals Central Ventilation System – Supply or Central Ventilation System – Exhaust and the “Total Installed Equivalent Continuous Ventilation” (C06) $\geq$ “Total Required Ventilation Rate” (B01), and if “Envelope Leakage”(D01) $\leq$ “Target dwelling unit compartmentalization leakage” (taken from NRCV-MCH-24), and the “Total Installed Equivalent Continuous Ventilation” (C06) $\leq$ “Maximum Ventilation Flow” (E01), then display text: “Building Passes Mechanical Ventilation Rate Test” Else display text: “Building Fails Mechanical Ventilation Rate Test”>>
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GK. Determination of HERSECC Verification Compliance

All applicable sections of this document shall indicate compliance with the specified verification protocol requirements in order for this Certificate of Verification as a whole to be determined to be in compliance.

01	<<if F01 = Building Passes Mechanical Ventilation Rate Test, then display: “Complies: All specified verification protocol requirements on this document are met”; else display: “Does not comply: One or more specified verification protocol requirements on this document are not met”>>
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